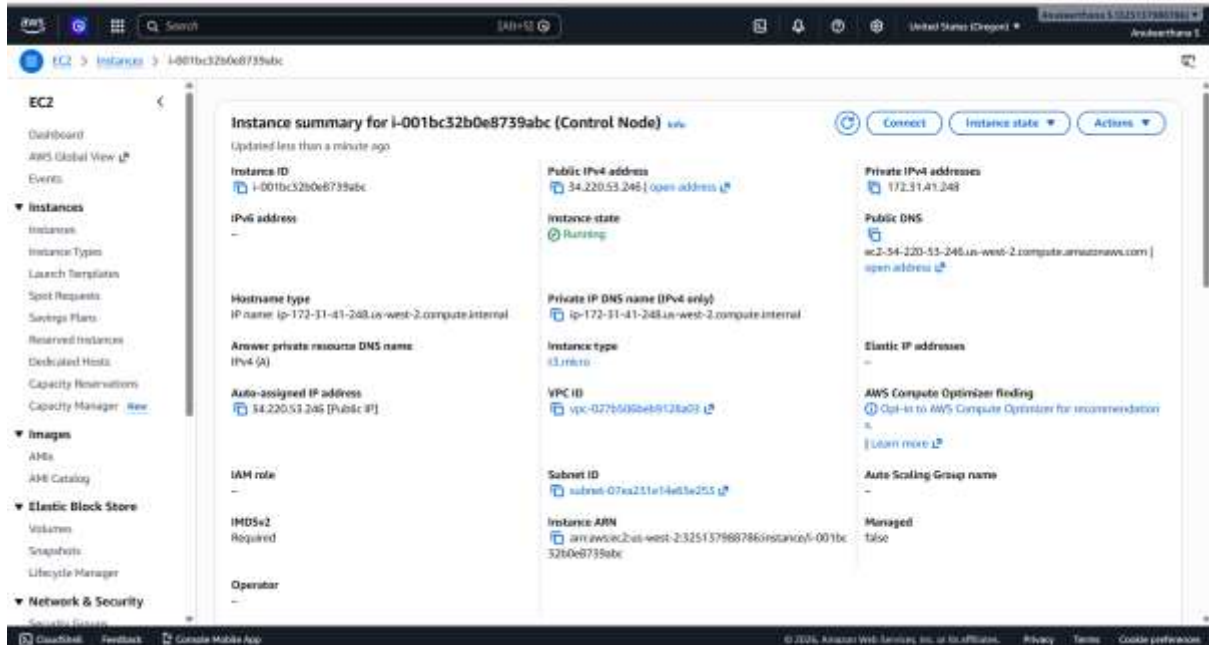


ANUKEERTHANA S

727723EUCS017

1.Control Node Configuration



2. Connect to Control Node

```
ubuntu@ip-172-31-41-248: ~  
C:\Users\Anukeerthana S>ssh -i "C:\Users\Anukeerthana S\Downloads\kp.pem" ubuntu@ec2-34-220-53-246.us-west-2.compute.amazonaws.com  
Warning: Identity file C:\Users\Anukeerthana S\Downloads\kp.pem not accessible: No such file or directory.  
ssh: Could not resolve hostname s\downloads\kp.pem: No such host is known.  
  
C:\Users\Anukeerthana S>cd Downloads  
  
C:\Users\Anukeerthana S\Downloads>ssh -i "kp.pem" ubuntu@ec2-34-220-53-246.us-west-2.compute.amazonaws.com  
The authenticity of host 'ec2-34-220-53-246.us-west-2.compute.amazonaws.com (34.220.53.246)' can't be established.  
ED25519 key fingerprint is SHA256:bFA8Q4/KTy3kpu/eeTx9ZPUfbcfL55sUzTK/GPo8kIs.  
This host key is known by the following other names/addresses:  
C:\Users\Anukeerthana S/.ssh/known_hosts:3: ec2-34-220-48-1.us-west-2.compute.amazonaws.com  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'ec2-34-220-53-246.us-west-2.compute.amazonaws.com' (ED25519) to the list of known hosts.  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1008-aws x86_64)  
  
 * Documentation:  https://docs.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:        https://ubuntu.com/pro  
  
System information as of Tue Jul 14 03:13:24 UTC 2026  
  
System load: 0.03      Temperature: -273.1 C  
Usage of /: 46.4% of 6.61GB  Processes: 120  
Memory usage: 25%      Users logged in: 0  
Swap usage: 0%         IPv4 address for ens5: 172.31.41.248  
  
Expanded Security Maintenance for Applications is not enabled.
```

3. Install Ansible on Control Node

```
ubuntu@ip-172-31-41-248: ~$ sudo apt update
Get:4 http://us-west-2.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [335 kB]
Get:5 http://us-west-2.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [87.9 kB]
Get:6 http://us-west-2.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [53.1 kB]
Get:7 http://us-west-2.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [284 kB]
Get:8 http://us-west-2.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [63.2 kB]
Get:9 http://us-west-2.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [153 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [277 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [72.7 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [31.2 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [133 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [41.9 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Fetched 1770 kB in 2s (1129 kB/s)
6 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-41-248:~$ sudo apt install ansible -y
ansible is already the newest version (13.1.0+dfsg-1ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 6
ubuntu@ip-172-31-41-248:~$ ansible --version
ansible [core 2.20.1]
  config file = None
  configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.14.4 (main, Jun 18 2026, 14:25:02) [GCC 15.2.0] (/usr/bin/python3)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
ubuntu@ip-172-31-41-248:~$
```

4. Configure SSH Key

```
Last login: Tue Jul 14 03:13:26 2026 from 117.239.103.162
ubuntu@ip-172-31-41-248:~$ ls
kp.pem
ubuntu@ip-172-31-41-248:~$ mkdir -p ~/.ssh
ubuntu@ip-172-31-41-248:~$ cp kp.pem ~/.ssh/aws-key.pem
ubuntu@ip-172-31-41-248:~$ chmod 400 ~/.ssh/aws-key.pem
ubuntu@ip-172-31-41-248:~$ ls -l ~/.ssh/aws-key.pem
-r----- 1 ubuntu ubuntu 1678 Jul 14 03:20 /home/ubuntu/.ssh/aws-key.pem
ubuntu@ip-172-31-41-248:~$
```

5. Verify Managed Node Python Availability

Managed Node 1

```

ssh: connect to host 172.31.44.4 port 22: Connection timed out
ubuntu@ip-172-31-41-248:~$ ssh -i ~/.ssh/aws-key.pem ubuntu@172.31.44.4
The authenticity of host '172.31.44.4 (172.31.44.4)' can't be established.
ED25519 key fingerprint is: SHA256:qhFB4dtSL+5f7PgYS2wWsXiOVg4caxUladj6ZV4MjZk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.31.44.4' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jul 14 03:28:15 UTC 2026

System load:  0.03          Temperature:   -273.1 C
Usage of /:   31.3% of 6.61GB Processes:      120
Memory usage: 26%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.44.4

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-44-4:~$ python3 --version
Python 3.14.4
ubuntu@ip-172-31-44-4:~$

```

Managed Node 2

```

ubuntu@ip-172-31-41-248:~$ ssh -i ~/.ssh/aws-key.pem ubuntu@172.31.40.173
The authenticity of host '172.31.40.173 (172.31.40.173)' can't be established.
ED25519 key fingerprint is: SHA256:/f8EoAEq4R75E+K9xgqf39NyETZF7MmAgdeYb/85RE4
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.31.40.173' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jul 14 03:30:25 UTC 2026

System load:  0.08           Temperature:   -273.1 C
Usage of /:   31.3% of 6.61GB Processes:     117
Memory usage: 26%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 172.31.40.173

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-40-173:~$ python --version
Command 'python' not found, did you mean:
  command 'python3' from deb python3
  command 'python' from deb python-is-python3
ubuntu@ip-172-31-40-173:~$ python3 --version
Python 3.14.4
ubuntu@ip-172-31-40-173:~$

```

6. Configure Ansible Inventory

```
ubuntu@ip-172-31-41-240: ~$ nano /etc/ansible/hosts
GNU nano 2.7.1 /etc/ansible/hosts
[webserver]

node1 ansible_host=172.31.44.4 ansible_user=ubuntu ansible_private_key_file=~/.ssh/aws-key.pem
node2 ansible_host=172.31.40.171 ansible_user=ubuntu ansible_private_key_file=~/.ssh/aws-key.pem

Help  Exit  Write Out  Read File  Where Is  Replace  Cut  Paste  Execute  Justify  Location  Go To Line  Undo  Redo  Set Mark  Copy  To Bracket  Where Was
```

7. Test Ansible Agentless Communication

```
ubuntu@ip-172-31-41-240:~$ sudo nano /etc/ansible/hosts
ubuntu@ip-172-31-41-240:~$ ansible all -m ping -i /etc/ansible/hosts
[WARNING]: Host 'node2' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause
a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.18/reference_appendices/interpreter_discovery.html for more information.
node2 | SUCCESS ==> {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.10"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host 'node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause
a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.18/reference_appendices/interpreter_discovery.html for more information.
node1 | SUCCESS ==> {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.10"
  },
  "changed": false,
  "ping": "pong"
}
ubuntu@ip-172-31-41-240:~$
```

8. Validate Idempotency


```
ubuntu@ip-172-31-42-246: ~$ cat test.yml
---
- name: Test Idempotency
  hosts: all
  become: yes

  tasks:
    - name: Ensure Python is installed
      apt:
        name: python3
        state: present
        update_cache: yes
```

```
ubuntu@ip-172-31-42-246:~$ ansible-playbook test.yml -i /etc/ansible/hosts

PLAY [Test Idempotency] *****

TASK [Gathering Facts] *****
[WARNING]: Host 'node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could run
as a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.10/reference_appendices/interpreter_discovery.html for more informa
tion.
ok: [node1]
[WARNING]: Host 'node2' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could run
as a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.10/reference_appendices/interpreter_discovery.html for more informa
tion.
ok: [node2]

TASK [Ensure Python is installed] *****
ok: [node1]
ok: [node2]

PLAY RECAP *****
node1                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
node2                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

ubuntu@ip-172-31-42-246:~$ ansible-playbook test.yml -i /etc/ansible/hosts

PLAY [Test Idempotency] *****

TASK [Gathering Facts] *****
[WARNING]: Host 'node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could run
as a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.10/reference_appendices/interpreter_discovery.html for more informa
tion.
ok: [node1]
[WARNING]: Host 'node2' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could run
as a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.10/reference_appendices/interpreter_discovery.html for more informa
tion.
ok: [node2]

TASK [Ensure Python is installed] *****
ok: [node1]
ok: [node2]

PLAY RECAP *****
node1                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
node2                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

ubuntu@ip-172-31-42-246:~$
```